Celebal Technologies Data Engineer Interview Handbook

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This handbook is a concise interview preparation guide covering the most important topics requested. It is intended as a revision companion.

SQL

GROUP BY groups rows for aggregate functions. HAVING filters grouped results. JOINs: INNER (matching), LEFT (all left), RIGHT (all right), FULL (all), CROSS (cartesian), SELF (table joined with itself). CTE (WITH) creates a temporary named result. Window functions compute values without collapsing rows: ROW_NUMBER, RANK, DENSE_RANK, LEAD, LAG.

Normalization

1NF: atomic values. 2NF: remove partial dependency. 3NF: remove transitive dependency. Denormalization intentionally adds redundancy for faster reads in analytics.

ACID

Atomicity, Consistency, Isolation, Durability ensure reliable transactions.

SCD

Type 1 overwrites old values. Type 2 preserves history by inserting new records with effective dates/current flag.

Medallion Architecture

Bronze: raw data. Silver: cleaned and validated. Gold: business-ready curated data.

Python

List: ordered mutable. Tuple: ordered immutable. Set: unordered unique values. Dictionary: key-value pairs. List comprehension: [x*x for x in range(5)]. OOP: Encapsulation, Inheritance, Polymorphism, Abstraction.

Important SQL Questions

1. Difference between WHERE and HAVING? Answer: WHERE filters rows before GROUP BY; HAVING filters groups after aggregation. 2. Difference between RANK and DENSE_RANK? Answer: RANK skips ranks after ties; DENSE_RANK does not. 3. Explain CTE. Answer: A temporary named result set defined using WITH, improving readability and supporting recursion. 4. Explain INNER vs LEFT JOIN. Answer: INNER returns matching rows only; LEFT returns all rows from left table and matching rows from right.

Important Python Questions

1. Why use tuples? Immutable and hashable. 2. What is list comprehension? Concise syntax for creating lists. 3. Explain inheritance. Child class acquires properties of parent class. 4. Difference between list and set? Lists allow duplicates and preserve order; sets store unique unordered elements.

Project Questions

Explain your ETL project end-to-end: Extract CSV -> Clean with Pandas -> Transform -> Load into PostgreSQL -> Visualize in Power BI. Explain Medallion Architecture project: Bronze->Silver->Gold using Azure Databricks and PySpark.